CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester/Second Semester of B. Tech. (All Branches) Examination

November-December 2013

ME 101 Engineering Graphics

Date: 06.12.2013, Friday   Time: 10:00 a.m. To 01:00 a.m.   Maximum Marks: 70

Instructions:

1. Attempt all the questions.
2. The question paper comprises of two sections.
3. Section I and II must be attempted in separate answer sheets.
4. Figures to the right indicate the full marks of the question
5. Figures drawn in the question paper are not to the scale.
6. Assume missing data or dimension and mention it clearly.
7. Take suitable scale whenever required and mention it clearly.
8. Use of scientific calculator is allowed.
9. Dimensions shown in the figure are in mm.
10. Show all the necessary dimensions in the solution.

SECTION-I

Q.1 (a) Answer the following statements with True or False (T/F). **[08]**

1. A Point is below HP and in front of VP; the point is in 4^{th} Quadrant.
2. In an isometric view, the angle of object is always 30^0.
3. Eccentricity of parabola is always 1.
4. Continuous thick line is the visible object outlines.
5. A line joining any point on spiral with the pole is called vectorial angle.
6. A line is parallel to HP and VP; the line is perpendicular to PP.
7. The RF = 1:1 is in Enlarge type of scale.
8. Cylinder has the infinite generators.

Q. 1(b) Define the representative fraction. **[02]**

Q.1 (c) The major axis and minor axis of an ellipse are 120 mm and 70mm long respectively. Draw an ellipse by any method. Draw normal and tangent at any point on the curve. **[10]**

OR

Q.1 (c) Two fixed straight lines OA and OB are at right angle to each other. A point P is at a distance of 40 mm from OA and 30 mm from OB. Draw a rectangular hyperbola passing through point P. **[10]**

Q. 2 A point P is 40 mm above H.P. & 40 mm in front of V.P. Draw the projections of point P. **[05]**

OR

Q. 2 A Point P is 30 mm above HP and is in first quadrant. Its shortest distance from XY line is 60 mm. Draw its plan and elevation. **[05]**

Q.3 Fig. 1 shows pictorial view of an object. Use 1^{st} angle projection method and draw (1) Front view in the direction 'X', (2) Top view. **[10]**

SECTION - II

Q.4 A line AB, 75mm long, has one end A in VP and other end B is 15 mm above HP. Draw the projections of the line when line is inclined 30° to HP and 60° to VP. **[10]**

OR

Q.4 A regular hexagonal plate 30 mm side is resting on one of its corners in HP. The diagonal through that corner is inclined at 30° to HP and the plan of that diagonal is inclined to VP by 45°. **[10]**

Q. 5 (a) A cone of base diameter 60 mm and height of 75 mm is resting on HP on its base in such a way that axis is inclined at 30° to HP and (i) the plan of axis is inclined at 45° to VP, (ii) axis is inclined at 45° to VP. Draw projection of the cone when apex is nearer to VP. **[10]**

OR

(a) Draw the development of square prism when it is rest on HP with its corner of base. Take the edge of base and axis height of 30 mm and 70 mm respectively. **[10]**

Q. 5 (b) Give the application of AutoCAD. **[05]**

OR

(b) Difference between Engineering Drawing and Engineering Graphics. **[05]**

Q.6 Fig. 2 shows the orthographic projections of an object in first angle projection system. Draw its isometric view. **[10]**

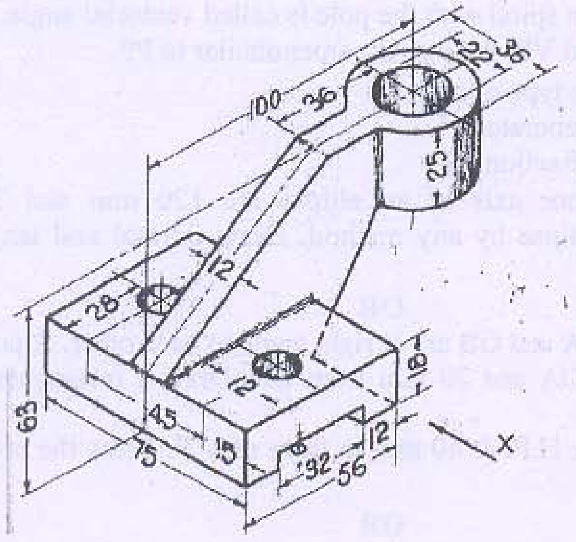

Figure 1

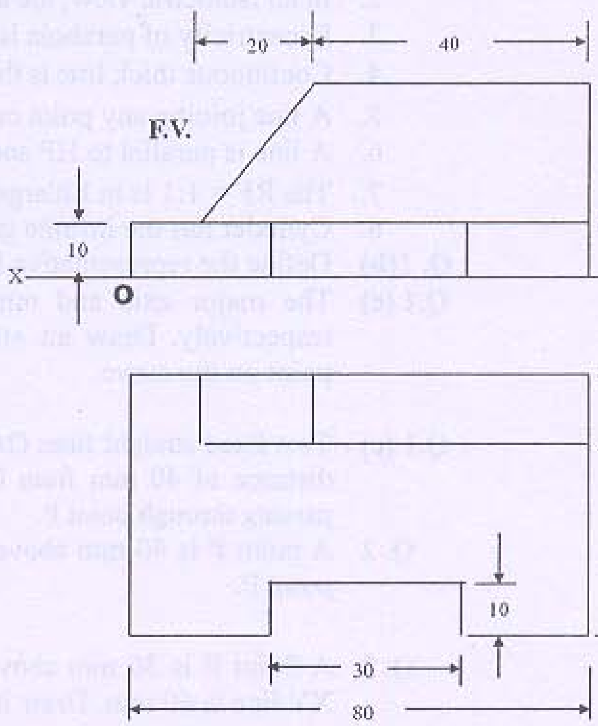

Figure 2